

Built-up Surface Extraction Pipeline

Tanzania Malaria Mapping — Non-Covariate Documentation

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Table of contents

Overview	2
1 Objectives	2
1.1 Non-Technical Summary	2
1.2 Technical Objectives	2
2 Data Sources	3
2.1 Primary Dataset — GHSL GHS-BUILT-S	3
2.2 Boundary Data — GADM / FAO GAUL	3
3 Epoch Mapping	4
4 Methods & Pipeline Steps	4
4.1 Environment Setup	4
4.2 GEE Initialization	5
4.3 Study Area Definition	5
4.4 Built-up Surface Extraction	5
4.5 Multi-band Stack Construction	6
4.6 Change Detection	6
4.7 Quality Control	7
5 Outputs	7
5.1 Directory Structure	7
5.2 Output File Descriptions	8
5.2.1 Separate Epoch Rasters (Output A)	8
5.2.2 Multi-band Stack (Output B)	8
5.2.3 Change Rasters (Output C)	8
6 Reproducibility	9
6.1 How to Reproduce	9
6.2 Scripts	9
7 Limitations	10

Overview

i For non-technical readers: This document describes how we extracted building footprint data across Tanzania — including Zanzibar and Pemba islands — to support malaria risk mapping. Building data helps us understand where people live in urban versus rural areas, which is strongly linked to malaria transmission patterns.

💡 For technical readers: This pipeline extracts GHSL GHS-BUILT-S P2023A built-up surface rasters for Tanzania (TZA) at 100m spatial resolution across epochs 2010, 2015, and 2020. Outputs include per-epoch rasters, a multi-band stack, and inter-epoch change rasters. The pipeline is implemented in Python using the Google Earth Engine Python API.

1 Objectives

1.1 Non-Technical Summary

Malaria transmission in Tanzania is strongly influenced by where people live. Urban areas typically have lower malaria risk because of better drainage and infrastructure, while peri-urban and rural areas often have higher risk due to more standing water and mosquito breeding sites.

To capture this pattern, we extracted **building footprint data** — a measure of how much of each area is covered by buildings — across all of Tanzania from 2010 to 2020. This gives us a way to distinguish:

- Dense urban centres (Dar es Salaam, Mwanza, Arusha)
- Peri-urban transition zones
- Rural areas including remote villages
- Island settlements on Zanzibar and Pemba

1.2 Technical Objectives

1. Extract gross built-up surface area (m² per 100m pixel) matching the definition of building footprint bounded by external wall perimeter at 1:10,000 topographic map specifications
2. Generate three temporal snapshots: **2010**, **2015**, and **2020**
3. Produce a multi-band raster stack for use in malaria modelling workflows
4. Quantify urban expansion between epochs via change detection rasters
5. Ensure full spatial coverage of Tanzania including Zanzibar and Pemba islands

2 Data Sources

2.1 Primary Dataset — GHSL GHS-BUILT-S

Attribute	Details
Full name	Global Human Settlement Layer — Built-up Surface (GHS-BUILT-S)
Version	P2023A (released 2023)
Producer	Joint Research Centre (JRC), European Commission
GEE Collection	JRC/GHSL/P2023A/GHS_BUILT_S
Band used	built_surface
Unit	m ² of gross building footprint area per 100m pixel
Native resolution	100m
CRS	EPSG:4326 (WGS84 Geographic)
Available epochs	1975, 1980, 1985, 1990, 1995, 2000, 2005, 2010, 2015, 2020
Epochs extracted	2010, 2015, 2020
Coverage	Global (full Tanzania including islands)
Access	Free via Google Earth Engine
Citation	Pesaresi et al. (2023). GHS-BUILT-S R2023A. European Commission, JRC.

! Important

Why GHSL GHS-BUILT-S? The `built_surface` band directly measures the gross building footprint area in m² per pixel — the area bounded by the external wall perimeter of all structures within each 100m cell. This exactly matches the project definition of built-up surface derived from 1:10K topographic map specifications.

2.2 Boundary Data — GADM / FAO GAUL

Attribute	Details
Source	FAO Global Administrative Unit Layers (GAUL) 2015
GEE Collection	FAO/GAUL/2015/level0, FAO/GAUL/2015/level1
Purpose	Clipping rasters to Tanzania national boundary

Attribute	Details
Coverage	Mainland Tanzania + Zanzibar (North, South, West) + Pemba (North, South)

3 Epoch Mapping

GHSL is produced in 5-year epochs. The project requested extraction from **2014 to present**. Since 2014 falls between the 2010 and 2015 epochs, the following mapping was used:

Requested Period	GHSL Epoch Used	Rationale
~2014 (baseline)	2010	Closest epoch before 2014; used as pre-period baseline
2015	2015	Exact match
2016–present	2020	Most recent complete epoch

i Note

The 2020 epoch represents the most recent built-up surface snapshot available in GHSL P2023A at the time of extraction. Future epochs will be incorporated as GHSL updates are released.

4 Methods & Pipeline Steps

4.1 Environment Setup

The pipeline runs in Python 3.10+ within an isolated virtual environment to ensure reproducibility.

```
# Create and activate virtual environment
python -m venv tz_malaria_env
tz_malaria_env\Scripts\activate      # Windows
# source tz_malaria_env/bin/activate # Mac/Linux

# Install dependencies
pip install earthengine-api geemap geopandas rasterio numpy matplotlib

# Save environment for reproducibility
pip freeze > requirements.txt

# Authenticate Google Earth Engine (once)
earthengine authenticate
```

Dependencies:

Package	Version	Purpose
earthengine-api	latest	GEE Python interface
geemap	latest	GEE image download
rasterio	1.3	Raster I/O and processing
numpy	1.24	Array operations
geopandas	latest	Vector boundary handling
matplotlib	3.7	Visualization

4.2 GEE Initialization

```
import ee

ee.Initialize(project="your-gee-project-id")
```

Google Earth Engine is used as the compute backend. Processing happens on Google's infrastructure — only the final exported rasters are downloaded locally.

4.3 Study Area Definition

Tanzania's national boundary is loaded directly from the FAO GAUL feature collection within GEE. This ensures the boundary includes:

- **Mainland Tanzania** — 945,087 km²
- **Zanzibar** — Unguja island (North, South, West regions)
- **Pemba** — Pemba island (North, South regions)

```
tz_national = (
    ee.FeatureCollection("FAO/GAUL/2015/level0")
    .filter(ee.Filter.eq("ADMO_NAME", "United Republic of Tanzania"))
)
aoi = tz_national.geometry()
```

4.4 Built-up Surface Extraction

For each epoch (2010, 2015, 2020):

1. Filter the GHS-BUILT-S image collection to the target epoch
2. Select the `built_surface` band
3. Clip to the Tanzania AOI
4. Clamp values to 0 (area cannot be negative)
5. Export to Google Drive as a GeoTIFF at 100m resolution
6. Download to local storage

```

builtup = (
    ee.ImageCollection("JRC/GHSL/P2023A/GHS_BUILT_S")
    .filter(ee.Filter.eq("system:index", str(epoch)))
    .first()
    .select("built_surface")
    .clip(aoi)
    .max(ee.Image(0))          # clamp negatives
)

```

Export parameters:

Parameter	Value
Scale	100m
CRS	EPSG:4326
Format	GeoTIFF (Cloud Optimised)
maxPixels	1×10^{13}

4.5 Multi-band Stack Construction

After downloading all epoch rasters, they are stacked into a single multi-band GeoTIFF using `rasterio`. Each band corresponds to one epoch:

Band	Name	Epoch
1	builtup_m2_2010	2010
2	builtup_m2_2015	2015
3	builtup_m2_2020	2020

This stack is the primary input for the malaria modelling covariate pipeline.

4.6 Change Detection

Built-up change between consecutive epochs is computed as simple pixel-wise subtraction:

$$\text{Change}_{t_1 \rightarrow t_2} = \text{BuiltUp}_{t_2} - \text{BuiltUp}_{t_1}$$

Value	Interpretation
Positive	New built-up area (urban expansion)
Zero	No change
Negative	Built-up area lost (demolition, reclassification)

Three change rasters are produced:

- 2010 → 2015 (5-year change)
- 2015 → 2020 (5-year change)
- 2010 → 2020 (full period change — primary output)

4.7 Quality Control

The following checks are applied to each raster:

Check	Threshold	Action
Negative pixel values	Any < 0	Clamped to 0
Pixel exceeds full area	$> 10,000 \text{ m}^2$	Warning flagged
Missing data coverage	Any NaN over Tanzania	Reported
Built pixel count	Plausibility check	Printed per epoch

Warning

Physical constraint: Each $100\text{m} \times 100\text{m}$ pixel has a maximum possible built-up area of $10,000 \text{ m}^2$ (if 100% of the pixel is covered by buildings). Any pixel value exceeding this threshold is flagged as a potential artefact.

5 Outputs

5.1 Directory Structure

```

outputs/tz_builtup/
  separate/
    builtup_ghsl_2010_100m.tif    ← Epoch 2010 (baseline ~2014)
    builtup_ghsl_2015_100m.tif    ← Epoch 2015
    builtup_ghsl_2020_100m.tif    ← Epoch 2020 (most recent)
  stack/
    tz_builtup_stack_2010_2015_2020_100m.tif  ← 3-band stack
  change/
    tz_builtup_change_2010_to_2015_100m.tif
    tz_builtup_change_2015_to_2020_100m.tif
    tz_builtup_change_2010_to_2020_100m.tif
  tz_builtup_visualization_2020.png  ← Visual QC map

```

5.2 Output File Descriptions

5.2.1 Separate Epoch Rasters (Output A)

File: `builtup_ghsl_<epoch>_100m.tif`

Property	Value
Format	GeoTIFF
Bands	1
Unit	m ² of gross building footprint per pixel
Resolution	100m
CRS	EPSG:4326
Compression	LZW
Data type	Float32
NoData	NaN
Coverage	Mainland Tanzania + Zanzibar + Pemba

5.2.2 Multi-band Stack (Output B)

File: `tz_builtup_stack_2010_2015_2020_100m.tif`

A single 3-band raster containing all epochs aligned to the same grid. Used as the primary covariate input for the malaria modelling pipeline.

Property	Value
Format	GeoTIFF
Bands	3 (one per epoch)
Band names	<code>builtup_m2_2010</code> , <code>builtup_m2_2015</code> , <code>builtup_m2_2020</code>
Unit	m ² of gross building footprint per pixel
Resolution	100m
CRS	EPSG:4326

5.2.3 Change Rasters (Output C)

File: `tz_builtup_change_<from>_to_<to>_100m.tif`

Pixel-wise difference between epochs showing urban expansion and loss.

Property	Value
Format	GeoTIFF
Bands	1
Unit	m ² change in building footprint per pixel
Positive values	New built-up area gained
Negative values	Built-up area lost
Zero values	No change

6 Reproducibility

6.1 How to Reproduce

1. Clone or download the project folder
2. Create the virtual environment:

```
python -m venv tz_malaria_env  
tz_malaria_env\Scripts\activate  
pip install -r requirements.txt
```

3. Authenticate GEE:

```
earthengine authenticate
```

4. Update the GEE project ID in the script:

```
ee.Initialize(project="your-gee-project-id")
```

5. Run the extraction:

```
python tz_builtup_extraction.py
```

6. Run visualization:

```
python tz_builtup_visualize.py
```

6.2 Scripts

Script	Purpose
tz_builtup_extraction.py	Main extraction pipeline (Steps 4–6)
tz_builtup_visualize.py	Visual QC map and histogram
requirements.txt	Frozen Python environment

7 Limitations

1. **Temporal resolution** — GHSL is produced in 5-year epochs; annual built-up data is not available at this time. Years between epochs are assigned to the nearest available epoch.
 2. **Epoch 2020 is the most recent** — built-up changes after 2020 are not captured until the next GHSL release.
 3. **Detection threshold** — GHSL GHS-BUILT-S may underestimate very small or informal structures (e.g., temporary shelters, single-room dwellings) common in rural Tanzania.
 4. **Zanzibar note** — informal settlement detection accuracy may be slightly lower on the islands due to fewer Sentinel-2 training samples relative to the mainland.
-

8 References

Pesaresi M., Politis P. (2023). GHS-BUILT-S R2023A - GHS built-up surface grid, derived from Sentinel-2 composite and Landsat, multitemporal (1975-2030). European Commission, Joint Research Centre (JRC). [doi:10.2905/9F06F36F-4B11-47EC-ABB0-4F8B7B1D72EA](https://doi.org/10.2905/9F06F36F-4B11-47EC-ABB0-4F8B7B1D72EA)

Google Earth Engine. (2024). JRC/GHSL/P2023A/GHS_BUILT_S. https://developers.google.com/earth-engine/datasets/catalog/JRC_GHSL_P2023A_GHS_BUILT_S

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